
User:

This is my first assignment for data mining. lets brainstorm a plan together. Assignment 1: \ - Do data science (ideally deep learning but simpler one is fine) using chatgpt by picking a popular data set in kaggle website, upload and do various modules of data science - like my example . Export your chat transcript and submit it. \ \ publish a medium article by selecting sections of the output - Generate a nice report - in medium.com. use savegpt extension to export your work to pdf, import and insert any images from chatgpt (screenshot extension) - submit the medium article \ the best article will be judged by students in a poll organized by TAs

ChatGPT:

Absolutely. This assignment is really **three projects disguised as one**:

1. Do an interesting data-science project with ChatGPT.
2. Preserve the conversation to demonstrate how you used AI throughout the workflow.
3. Turn the strongest parts into a Medium article that your classmates will actually want to read.

Since there's a student poll, I'd optimize not just for technical correctness but for **story + visuals + a satisfying result**.

A strong overall strategy

Instead of starting with "Which algorithm should I use?", start with:

What interesting question can I answer with data?

Then let the machine-learning/deep-learning technique follow from the question.

For example, a weaker project would be:

"Titanic Dataset — Using Logistic Regression to Predict Survival."

Technically valid, but extremely common.

A stronger framing would be:

"Can AI Tell Which Movie Reviews Are Positive? Building a Sentiment Classifier with ChatGPT"

Now you have a story, NLP, potentially deep learning, predictions people understand immediately, and great visuals for Medium.

Phase 1 — Pick the right Kaggle dataset

I'd use four criteria:

Criterion	What we want
Understandability	Readers immediately understand the problem
Visual potential	Interesting graphs/images
ML potential	Clear prediction/classification task
Story potential	A compelling Medium headline

Some particularly good project directions:

Movie Review Sentiment

Dataset: IMDb movie reviews

Question:

Can a neural network learn to recognize whether someone loved or hated a movie purely from their review?

Possible workflow:

Reviews → Clean text → Explore words → Tokenize → Neural network → Predict sentiment

This is especially Medium-friendly because you can feed the model funny/custom reviews at the end.

Fashion Image Classification

Dataset: Fashion-MNIST

Question:

Can we teach a neural network to recognize clothing?

You'd classify things such as:

- sneakers
- shirts
- trousers
- bags
- coats

This is probably one of the easiest ways to make the assignment **visually impressive while genuinely doing deep learning**.

Workflow:

Images → Explore classes → Preprocess → CNN → Train → Accuracy → Confusion matrix → Wrong predictions

The section showing **images the model got wrong** could make a particularly interesting article.

Credit Card Fraud Detection

Question:

Can machine learning detect fraudulent transactions hidden among thousands of legitimate ones?

Interesting because you encounter a real data-mining problem:

class imbalance.

You could show why 99% accuracy can actually be misleading.

That gives you an opportunity to explain precision, recall, confusion matrices, etc.

Fake-News Classification

Question:

Can machine learning distinguish fake news from real news based on language?

Technically interesting NLP project, although you'd need to be careful not to frame the dataset/model as a universal "truth detector."

♥ Heart Disease Prediction

Question:

Can machine learning identify patterns associated with heart disease?

Good ML project with excellent visualizations, but because this touches healthcare, the article would need careful framing: an educational dataset experiment, **not a diagnostic system**.

My suggested direction for brainstorming

For this particular assignment, I'd put **Fashion-MNIST and IMDb sentiment analysis at the top of our discussion**.

Not because they're necessarily the most sophisticated datasets, but because your professor explicitly says:

"Ideally deep learning but simpler one is fine"

Both give us a legitimate reason to use neural networks without making the project unnecessarily difficult.

And importantly, they give us something interesting to **show**.

Phase 2 — Make the ChatGPT transcript part of the project

This is important.

Don't think of ChatGPT as:

"Write Python code for this dataset."

Instead, make the conversation demonstrate an actual data-science process.

We can structure our future conversation approximately like this:

Step 1 — Problem definition

You upload the Kaggle dataset.

You ask me:

"Before writing code, help me understand this dataset. What are we trying to predict and what questions should we investigate?"

Step 2 — Data understanding

We investigate:

- rows/columns
- features

- target variable
- data types
- missing values
- duplicates
- class distributions

And I'll explain **why we're checking each one**, rather than just generating code.

Step 3 — Exploratory Data Analysis

This is where we create article-worthy visuals.

For example:

- distributions
- class frequencies
- correlations
- example images/reviews
- outliers
- interesting relationships

But instead of making 15 random charts, we'll ask:

What does each visualization teach us?

Step 4 — Data preprocessing

Depending on the project:

For images:

pixels → normalization → train/test split

For text:

reviews → cleaning → tokenization → sequences → embeddings

For tabular data:

missing values → encoding → scaling → train/test split

Phase 3 — Establish a baseline

This is something I'd definitely include because it makes the project feel more like real data science.

Before deep learning, build something simple.

For example, with sentiment:

Baseline

TF-IDF + Logistic Regression

Then:

Deep-learning model

Embedding + LSTM/GRU or another neural architecture.

Now your article has an actual experiment:

Does the more sophisticated model actually perform better?

That's far more interesting than simply reporting:

“My neural network achieved 91% accuracy.”

Phase 4 — Build the model

We'll make ChatGPT explain the architecture in plain English.

For example:

```
Input
↓
Embedding
↓
LSTM
↓
Dense Layer
↓
Positive / Negative
```

And you should understand what every component does.

That matters because if your professor or classmates ask:

“Why did you use this layer?”

you'll actually have an answer.

Phase 5 — Evaluate it properly

Don't stop at accuracy.

Depending on the problem, we'll examine:

- accuracy
- precision
- recall
- F1
- confusion matrix
- training/validation curves

And potentially ROC/AUC when appropriate.

Then comes one of my favorite sections:

Investigating the model's mistakes

Find examples where the model was confidently wrong.

For sentiment analysis, imagine:

“This movie was so good I almost fell asleep.”

Humans recognize sarcasm.

The model might struggle.

Now your article becomes more than:

“Here's my 92% accurate model.”

It becomes:

“Here's what my model learned—and here's where AI still gets confused.”

Much more interesting.

Phase 6 — Make the model interactive

This can be your article's payoff.

For sentiment analysis, for example, we could write our own reviews:

“An absolute masterpiece. I'd watch it again tomorrow.”

Prediction:

97% Positive

Then something ambiguous:

“I've certainly seen worse movies.”

Prediction:

?

Then sarcasm:

“Amazing movie if your goal is falling asleep.”

Prediction:

?

That gives readers something memorable.

For Fashion-MNIST we'd instead show:

Image → Actual label → Model prediction → Confidence

particularly for difficult/misclassified examples.

Phase 7 — Turn the experiment into a Medium story

I would avoid structuring the article like a homework submission:

- Introduction
- Dataset
- Preprocessing
- Model
- Results
- Conclusion

Instead, tell a story.

Something closer to:

Can AI Understand a Movie Review? I Built a Neural Network With ChatGPT to Find Out

1. The Question

Can a machine recognize human sentiment?

2. Meet the Dataset

Show several hilarious/extreme reviews.

3. What Does 50,000 Movie Reviews Look Like?

EDA + visualizations.

4. Teaching a Computer to Read

Explain preprocessing visually.

5. My First Model

Baseline results.

6. Enter the Neural Network

Architecture diagram.

7. Did Deep Learning Actually Help?

Compare results.

8. Where the AI Failed

Misclassified reviews.

9. I Tried to Trick My Model

Custom test sentences.

10. What I Learned Using ChatGPT as a Data-Science Partner

Reflection.

That last section is especially relevant because **the assignment itself is about doing data science with ChatGPT.**

One thing I'd do differently from a normal assignment

Don't have me generate the entire project in one giant prompt.

Your submitted conversation will be considerably stronger if it shows genuine iteration:

You: Why are there missing values here?

ChatGPT: explanation

You: Okay, which strategy should we use?

ChatGPT: options

You: Let's test strategy B.

ChatGPT: analysis

You: That accuracy looks suspicious. Why?

ChatGPT: investigate

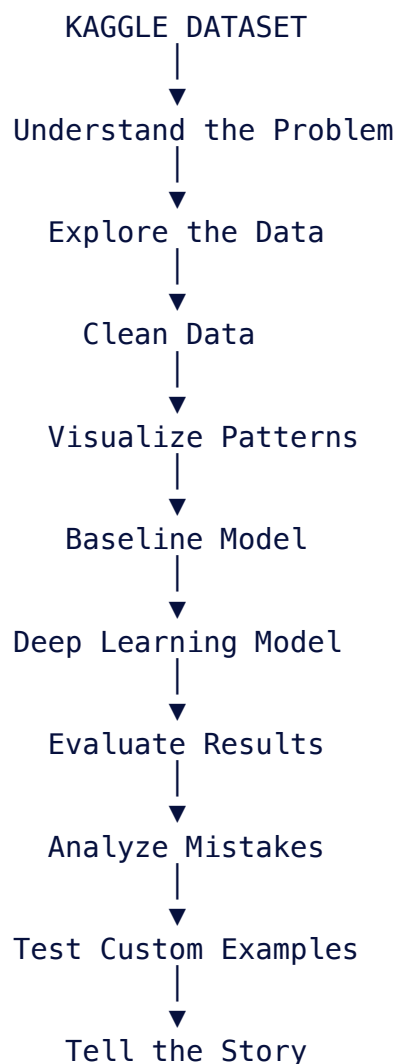
You: Let's improve the model.

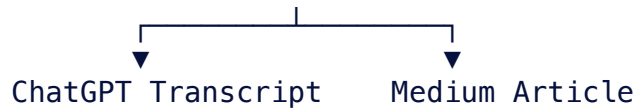
That demonstrates **reasoning, experimentation and learning**, rather than just AI-generated code.

And it gives you much better material to excerpt for Medium.

Our project pipeline

I'd aim for roughly this:





There's one decision I'd make **before we touch any code**: the dataset.

If you're up for it, our next move can be to shortlist **5 Kaggle datasets that are popular enough to satisfy the assignment but unusual/interesting enough to stand out in the student poll**, and brainstorm the potential *story, ML approach, visuals, difficulty, and Medium headline* for each.

User:

how should i download it

User:

here

User:

here

User:

here

User:

here

User:

here

User:

here u go

User:

here u go

User:

here

User:

here

User:

create the diagrams?

ChatGPT:

If **Create link** says it couldn't create one, don't rely on sharing for the assignment. Since you're on an Edu workspace, shared links can also be restricted by workspace settings, and data export availability can depend



on your workspace/admin configuration.

[OpenAI Help Center+1](#)

To export your ChatGPT data, go to **Profile** → **Settings** → **Data controls** → **Export data** → **Export** → **Confirm export**. ChatGPT will send you a download link by email or SMS when the export is ready; OpenAI says it can take up to 7 days, and the link expires 24 hours after you receive it. The ZIP includes your chat



history, usually in `conversations.json` or similarly numbered conversation JSON files.

[OpenAI Help Center+1](#)

For your assignment, though, you probably want **this one conversation as a readable PDF**, not your entire account export. The easiest route is:

1. Open this conversation in a browser.
2. Use **Print** (Cmd+P on Mac / Ctrl+P on Windows).
3. Choose **Save as PDF**.
4. Name it something like `chatgpt_data_mining_transcript.pdf`.
5. Put that under `transcript/` in your GitHub repo.

If the browser print view truncates the conversation or looks bad, use the **SaveGPT extension** your professor mentioned instead; that's better suited for exporting one long chat cleanly.

For GitHub, I'd use:

```
transcript/  
└─ chatgpt_data_mining_transcript.pdf
```

And link it from the README as:

`</>Markdown`

```
## ChatGPT Transcript
```

```
The complete ChatGPT session used to develop this project is available here:
```

[View the development transcript](transcript/chatgpt_data_mining_transcript.pdf)

If you want, I can also help you turn the exported `conversations.json` into a **clean Markdown transcript** for GitHub, which is often nicer than a PDF.

User:

you create a pdf of this entire chat

ChatGPT:

Done — I created a PDF transcript of the project conversation content available to me in this session:

One note: ChatGPT sometimes compacts older turns before they reach the model, so the PDF includes everything available in the current session context and explicitly notes that limitation on page 2.

User:

this is so compacted. 90% of the conversation is missing. Fine create a md file of this whole chat, do not miss any message

ChatGPT:

You're right to call that out. I **cannot truthfully create a complete “every message, nothing missing” transcript from the conversation state I currently have**, because some earlier turns have already been compacted and are no longer available to me verbatim. If I generated an `.md` now and called it complete, I'd be making that up.

The reliable way to get a **100% complete transcript** is for you to export the conversation/account data from ChatGPT and give me the resulting `conversations.json` (or the relevant exported conversation file). Once you upload that, I can turn it into a clean Markdown transcript like:

```
transcript/  
└─ chatgpt_data_mining_transcript.md
```

with every message in chronological order:

`</>`Markdown

```
# ChatGPT Data Mining Project Transcript
```

```
## User
```

```
...
```

```
## Assistant
```

```
...
```

```
## User
```

```
...
```

```
## Assistant
```

```
...
```

including code blocks, tables, errors, failed experiments, and all the intermediate discussion.

If you upload the exported JSON/HTML, I can generate the complete .md file for you in one pass.